



# Disulfide connectivity prediction based on structural information without a prior knowledge of the bonding state of cysteines



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## ABSTRACT

Previous studies predicted the disulfide bonding patterns of cysteines using a prior knowledge of their bonding states. In this study, we propose a method that is based on the ensemble support vector machine (SVM), with the structural features of cysteines extracted without any prior knowledge of their bonding states. This method is useful for improving the predictive performance of disulfide bonding patterns. For comparison, the proposed method was tested with the same dataset SPX that was adopted in previous studies. The experimental results demonstrate that bridge classification and disulfide connectivity predictions achieve 96.5% and 89.2% accuracy, respectively, using the ensemble SVM model, which outperforms the traditional method (51.5% and 51.0%, respectively) and the model that is based on a single-kernel SVM classifier (94.6% and 84.4%, respectively). For protein chain and residue classifications, the sensitivity, specificity, and accuracy of ensemble and single-kernel SVM approaches are better than those of the traditional methods. The predictive performances of the ensemble SVM and single-kernel models are identical, indicating that the ensemble model can converge to the single-kernel model for some applications.

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## 1. Introduction

Disulfide bonds constitute an important cross-linkage between cysteine side chains in proteins and are known to play a key role in stabilizing protein conformations and functions. Protein folding simulations demonstrate that correctly predicted disulfide bonding patterns can efficiently reduce the search space [1,2]. The disulfide bonds impose geometrical constraints on the protein backbones; therefore, the correct prediction of the disulfide bonding pattern may greatly help to predict the three-dimensional structure of a protein, which, in turn, can manifest its function.

Disulfide bonding patterns can be divided into inter- and intra-chain disulfide bonds. Niu et al. [3] provided a method for the classification of inter- and intra-chain disulfide bonds. In previous works, the prediction of disulfide bonding patterns only focused on intra-chain disulfide bonds because the cysteines that are contained in inter-chain disulfide bonds are considered free cysteines. The realm of disulfide bond predictions can be divided into four problems [4]: (i) protein chain classification to determine whether a protein contains disulfide bridges; (ii) residue classification to categorize the bonding state of cysteines; (iii) bridge classification to predict whether a pair of cysteines is linked by a disulfide bond; and (iv) disulfide bonding pattern prediction to predict the cysteine pairs that are bonded to each other.

Recently, several approaches have been proposed to predict cysteine bonding states and disulfide bonding patterns. These approaches can be grouped into the following three categories: (i) methods for predicting the cysteine bonding states [5–8]; (ii) approaches for predicting the disulfide bonding patterns with a prior knowledge of the cysteine bonding states [4,9–17]; and (iii) methods for predicting both cysteine bonding states and

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disulfide bonding patterns [4,11]. In the prediction of cysteine bonding states, several methods that are based on statistical analysis [5], neural networks [6,7], and support vector machines [8] have been proposed, with significant progress being made in the prediction of the cysteine bonding state, with achieved accuracies ranging from 81% to 90%. The computational approaches that are used to predict disulfide bonding patterns have also been proposed recently. Fariselli and Casadio [9] proposed a method to convert the prediction problem into a graph matching problem with vertices that indicate the oxidized cysteines and edge weights that are labeled as the contact potentials between the corresponding pairs of cysteines. The optimal values of contact potentials were obtained using the Monte Carlo simulated annealing method, and then the disulfide bonds were located by finding the maximum weight perfect matching. Vullo and Frascioni [10] applied an ad-hoc recursive neural network to improve the prediction accuracy from 34% to 44%. Cheng et al. [4] used two-dimensional recursive neural networks for predicting the connectivity probabilities between cysteine pairs to further improve the accuracy. Ferrè and Clote [11] used the secondary structure information and diresidue frequency to train the predictive model that was designed based on the diresidue neural network to predict connectivity probabilities between cysteine pairs. Tsai et al. [12] used the local sequence profiles and the linear distance of cysteines as the features for training the support vector machine (SVM) model to predict connectivity probabilities between cysteine pairs.

In contrast to the aforementioned approaches that were based on the conversion of a disulfide bonding pattern prediction problem to a maximum weight perfect matching problem, Cheng and Hwang [13] directly predicted the disulfide bonding patterns by a model that was designed with the SVM model, which was based on the following features: coupling between the local sequence environments of cysteine pairs, cysteine separations, and amino acid contents. Conversely, Zhao et al. [14] used a simple feature, i.e. cysteine separation profiles (CSPs), which is based on the assumption that similar protein structures have similar disulfide bonding patterns, to predict the disulfide bonding pattern. Chen et al. [15] proposed a two-level model, consisting of a pair-wise level and a pattern-wise level, by extending local information regarding cysteine pairs (pair-wise) to global information, including protein length, cysteine separation, and disulfide connectivity frequency (pattern-wise) for the prediction of disulfide bonding patterns. Lu et al. [16] adopted the genetic algorithm (GA) to optimize the feature selection for training the SVM model, achieving an accuracy of 73.9%. Song et al. [17] used the multiple sequence vectors and secondary structure information to train a support vector regression model for the prediction of disulfide bonding patterns, with an accuracy of 74.4%. Recently, Lin and Tseng [18] used features, including position-specific scoring matrix (PSSM), normalized bond lengths, predicted secondary structure of proteins, and indices of the physicochemical properties of amino acids for training a SVM model for the prediction of disulfide bonding pattern with an accuracy of 79.8%.

The computational methods for the prediction of disulfide bonding patterns can be divided into sequence-based and structural-based [19]. In general, the features used in the aforementioned methods are primarily extracted from the protein sequence. In this study, the structural information, i.e. X, Y, and Z coordinates of the  $C_\alpha$  of each amino acid, which was contained in the protein that was predicted by the MODELLER software [20] was used to calculate the feature *NPD*. In the work that was performed by Lin and Tsang [21], a single-kernel SVM was used for the bridge classification and prediction of disulfide bonding patterns, which mainly focused on the web service to provide a tool for biologists and medical scientists. In contrast, this study applied an ensemble

SVM consisting of three kernels to train the model to compute the connectivity probabilities of cysteine pairs, followed by the modified maximum weight perfect matching algorithm to find the disulfide bonding pattern. This SVM has been demonstrated to have better predictive performance than the methods that were proposed by Cheng et al. [4] and the single-kernel SVM model [21]. The method proposed in this work aims to improve the predictive performance of the disulfide bonding pattern of a protein sequence that does not have cysteines involved in the metal-binding sites. An updated version of the web service [21] is available at <http://biomedical.ctust.edu.tw/edbcp>.

## 2. Materials and methods

### 2.1. Datasets

To compare the predictive performance of our proposed method with previously reported methods [4,7,8], the same dataset, which was denoted as MART and provided by Martelli et al. [7], was employed for the experiments. In the dataset MART, a total of 4136 segments containing cysteines in 969 protein sequences were extracted from PDB [22] with sequence identities less than 25% and without chain breaks (non-homologous). The segments with cysteines that were inter-chain disulfide bonded were included as ‘free’ cysteines (non-disulfide-bonded). Among the 4136 segments with cysteines, 2690 were in the free state, and the other 1446 were in the disulfide bonded state. The dataset was split into 20 subsets of roughly equal size by Martelli et al. for 20-fold cross-validation to verify the performance of their proposed method.

The dataset SPXC was adopted to address the problem of protein chain classification, whereas the dataset SPX was used to pinpoint the problems regarding residue classification, bridge classification, and disulfide bonding pattern prediction. The datasets SPXC and SPX were prepared by Cheng et al. [4], with all the proteins in both SPXC and SPX datasets that were extracted from the PDB on May 17, 2004. To remove overrepresentation of particular protein families, the UniqueProt tool [23], which was designed based on the HSSP distance [24] to reduce protein redundancy, was adopted [4]. In dataset SPXC, there are 897 positive sequences that were selected with an HSSP cutoff distance of 5 and 1650 negative sequences that contained no disulfide bridges with an HSSP cutoff distance of 0. The dataset SPX is a collection of 1018 proteins, which contain at least one intra-chain disulfide bond and at least 12 amino acids that were obtained by setting the HSSP cutoff distance to 10. To compare the methods that were proposed in this study and the method that was proposed by Cheng et al. [4], the protein sequences were randomly divided into 10 subsets of roughly equal size for 10-fold cross-validation.

### 2.2. Methodology

The method proposed by Cheng et al. [4] can be divided into two stages. This method first predicted the bonding state of cysteines, and then oxidized cysteines were used for the prediction of disulfide bonding patterns. In this study, in contrast, the bonding probability of all the cysteine pairs was directly predicted. The normalized pair distance (*NPD*) was adopted as the feature, and then the SVM model was trained to compute the connectivity probabilities of cysteine pairs. Afterward, the MTS [25] was used to evolve the SVM parameters, i.e.,  $C$  and  $\gamma$ , the *NPD* window sizes, and the weights of individual models in the ensemble SVM classifier. Finally, the modified maximum weight perfect matching algorithm was then used to find the disulfide connectivity pattern without a prior knowledge of the bonding state of cysteines.

### 2.3. Normalized pair distance (NPD)

With the exception of the protein's secondary structure, the features that were used in the previous studies for predicting disulfide bonding pattern include the ones that were extracted from the protein sequence, which are unrelated to the protein structure. In this study, the structure-related feature, *NPD*, was adopted for the prediction of the disulfide bonding pattern. The MODELLER software [20] was used to predict the coordinates (*X*, *Y*, and *Z*) of the  $C_\alpha$  of each amino acid in the protein sequence. With these coordinates, the Euclidean distance  $D_{ij}$  between the amino acids at the *i*th position and at the *j*th position can be computed. We further extended the definition of the Euclidean distance to the pair distance (*PD*). Let the positions of cysteines *i* and *j* be  $P_i$  and  $P_j$ , respectively. The *PD* between cysteines *i* and *j* is defined as a vector  $(D_{P_i-[L/2],P_j-[L/2]}, \dots, D_{P_i-1,P_j-1}, D_{P_i,P_j}, D_{P_i+1,P_j+1}, \dots, D_{P_i+[(L-1)/2],P_j+[(L-1)/2]})$  containing *L* Euclidean distances with *L* indicating the sliding window size of the *NPD* feature [21]. If there are *k* cysteines existing in the protein, then as many as  $k(k-1)/2$  cysteine pairs can be obtained. By examining  $D_{P_i,P_j}$  of the cysteine pairs, a threshold value of 15 has been set for determining the disulfide bond candidate pairs because most cysteine pairs contain no disulfide bonds. If  $D_{P_i,P_j}$  is greater than 15, then this cysteine pair will not be considered as a disulfide bond candidate. To adjust the feature values to lie within the range from  $-1$  to  $1$  that are

suitable to input into the SVM model, each component of the *PD* vector was normalized according to the equation  $(D_{P_i,P_j} - 7.5)/7.5$ , resulting in a new vector named normalized pair distance (*NPD*).

### 2.4. Multiple trajectory search

The multiple trajectory search (MTS) learning method was proposed for solving large-scale global optimization problems [25]. MTS has also been used to solve multi-objective optimization problems with satisfactory results [26]. In MTS, multiple agents were used to concurrently search the solution space, in which each agent performed an iterated local search using one of three candidate methods for local searching. By choosing one of the local search methods that best fits the landscape of a solution's neighborhood, an agent may find its way to a local optimum or to the global optimum.

### 2.5. Ensemble classifier

SVM with a single-kernel has been widely applied in several research areas, such as biomedical, clinic decision support systems, bioinformatics and pattern recognition. An ensemble of classifiers is a combination of multiple classifiers. An ensemble classifier often obtains better performance and is more stable than an

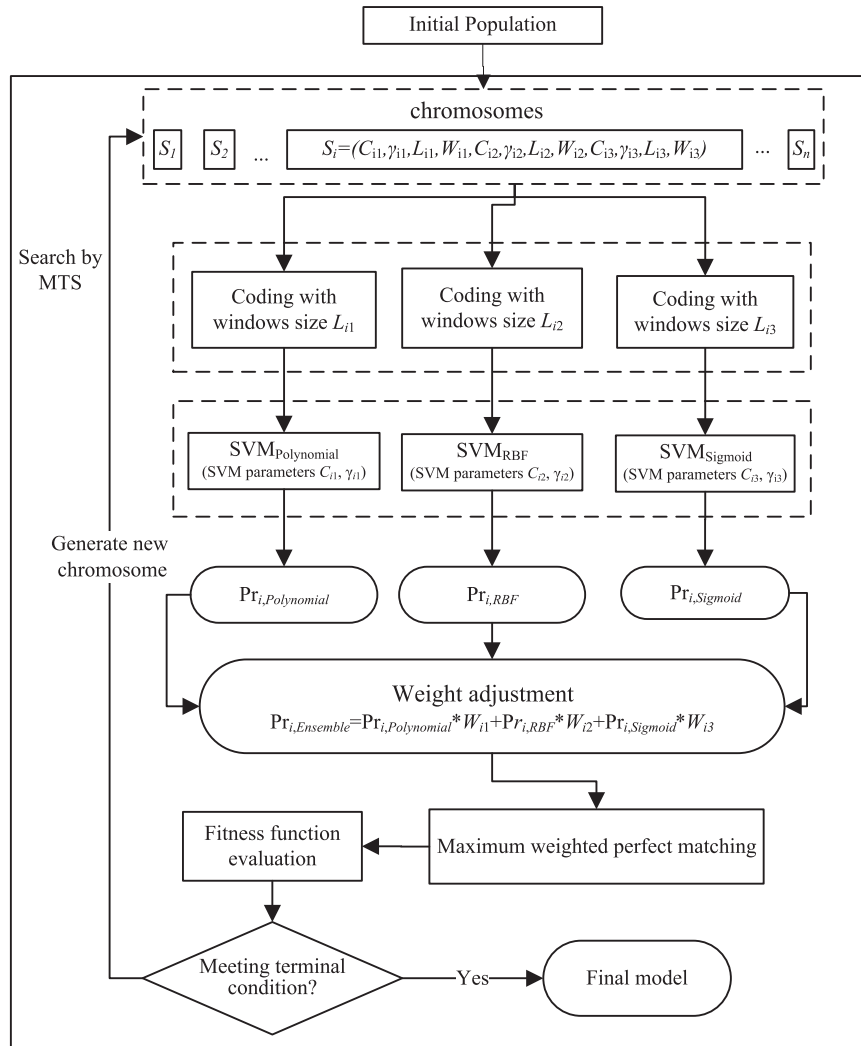


Fig. 1. Flowchart of the proposed method to integrate MTS and the ensemble SVM model.

individual constituent classifier [27–29]. In this study, the ensemble model, which consisted of three SVM models, each with a kernel, i.e., polynomial, RBF, and sigmoid, and with its own weight proportional to its significance, was used for bridge classification and the prediction of the disulfide bonding pattern.

## 2.6. Multiple trajectory search for selecting SVM parameters, window sizes, and probability weights

It is widely believed that SVM is superior to the traditional statistical and neural network classifiers in many applications. The LIBSVM tool [31] was used to train the SVM model in this study. The values of SVM parameters, i.e.,  $C$  and  $\gamma$ , are critical in optimizing classification performance. Traditionally, a regular grid search strategy was performed to select a combination of parameters with best performance; however, it is time-consuming. In this study, the MTS and the ensemble SVM model were tightly integrated for training the model with more efficiency. A chromosome for MTS is defined as  $S_i = (C_{i1}, \gamma_{i1}, L_{i1}, W_{i1}, C_{i2}, \gamma_{i2}, L_{i2}, W_{i2}, C_{i3}, \gamma_{i3}, L_{i3}, W_{i3})$ , in which  $C_{ij}$  and  $\gamma_{ij}$  indicate the log values of the SVM parameters,  $L_{ij}$  is the window size of  $NPD$ , and  $W_{ij}$  denotes the probability weight of  $j$ th model that is embedded in the ensemble SVM classifier with  $j=1, 2$ , and  $3$  representing the polynomial, RBF, and sigmoid kernels, respectively. The ensemble probability of all

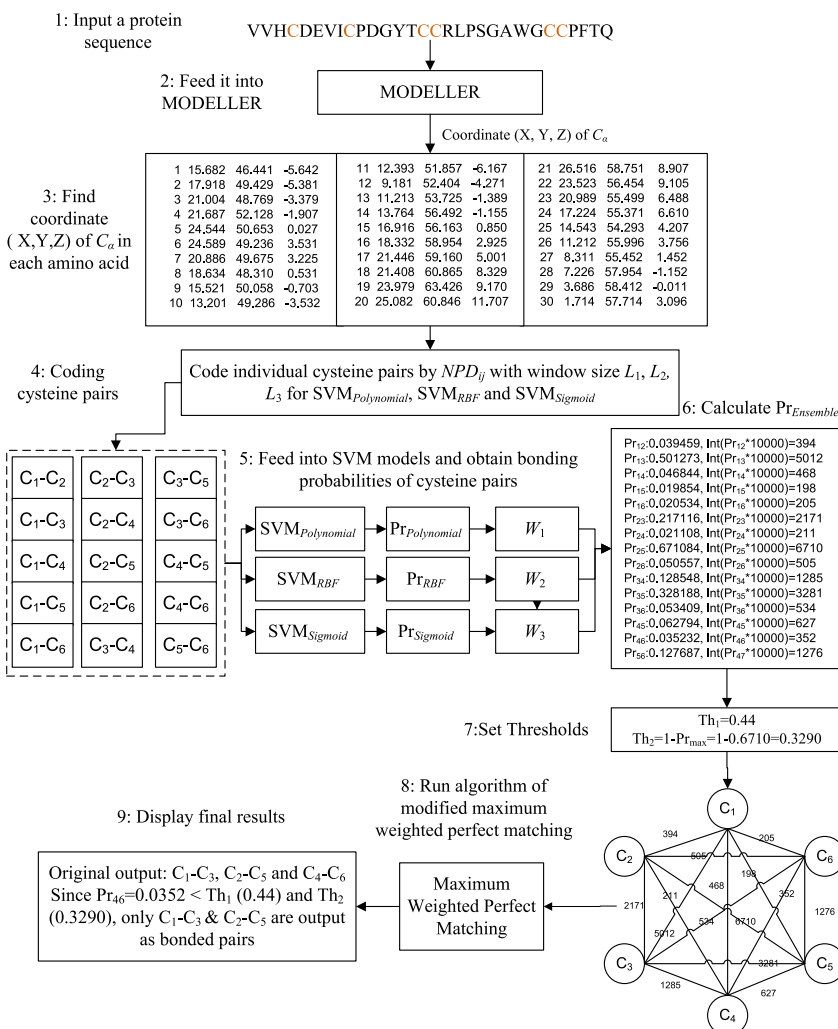
the cysteine pairs can be derived by the following equation:

$$\Pr_{i,Ensemble} = (\Pr_{i,Polynomial} \times W_{i1} + \Pr_{i,RBF} \times W_{i2} + \Pr_{i,Sigmoid} \times W_{i3}) / (W_{i1} + W_{i2} + W_{i3}) \quad (1)$$

The fitness function is defined as the accuracy of the ensemble SVM model with regards to disulfide connectivity prediction. The flowchart of the proposed method which integrates the MTS and the SVM models is shown in Fig. 1.

## 2.7. Modified maximum weight perfect matching

When a test protein is given, the ensemble probabilities of all the cysteine pairs can be obtained by the ensemble SVM model. As stated before, if  $D_{pi,pj}$  is greater than 15, then it is assumed that there is no bond between cysteine  $i$  and cysteine  $j$ ; therefore, the bonding probability  $\Pr_{ij}$  is set to 0. After the bonding probabilities of the candidate cysteine pairs have been found, Gabow's algorithm [32] is then applied to find the maximum weighted perfect matching. The Gabow's algorithm is suitable to find the bonding pattern, especially when all the cysteines are bonded. If there are  $B$  bonded cysteines observed in a protein, then exactly  $B/2$  pairs can be found. However, for predicting the disulfide bonding pattern without a prior knowledge of the bonding state of cysteines, Gabow's algorithm will output only  $\lfloor k/2 \rfloor$  cysteine pairs for  $k$  cysteines in a protein. In reality, among the  $k$  cysteines in a protein,



some could be unbonded. This observation indicates that the output of Gabow's algorithm may contain unbonded cysteine pairs.

To solve this problem, two probability thresholds were used to determine whether a bond exists between two cysteines. In this study, the first threshold ( $Th_1$ ) was set to 0.44, which was obtained from the training phase. Nevertheless, among the output cysteine pairs for some proteins after applying Gabow's algorithm, the probabilities of some cysteine pairs were higher (such as 0.9), whereas other cysteine pairs had lower probabilities (such as 0.2). According to the first threshold, the cysteine pairs with probabilities that were higher than  $Th_1$  were deemed as bonded. To increase the predictive performance, the second threshold ( $Th_2$ ), which was defined as  $1 - Pr_{max}$ , with  $Pr_{max}$  indicating the maximum probabilities of all the cysteine pairs, was applied to remove false negative cases.

### 2.8. An example illustrating the procedure of the proposed method

Fig. 2 illustrates an example that presents the procedure of the proposed ensemble SVM model. As shown in the figure, a protein sequence containing 30 amino acids with six cysteines, is input (Step 1) for generating their corresponding coordinates of  $C_\alpha$  by the MODELLER software (Steps 2 and 3). In Step 4, the 15 cysteine pairs that were obtained from six cysteines are then coded by *NPD* features with three individual window sizes, i.e.  $L_1$ ,  $L_2$ , and  $L_3$ , for  $SVM_{Polynomial}$ ,  $SVM_{RBF}$  and  $SVM_{Sigmoid}$ , respectively. The data of 15 cysteine pairs are then input to train and test the ensemble SVM model for obtaining the model with the best performance (Step 5). After training and testing on the dataset SPX, the resulting parameter values of  $\log_2 C$ ,  $\log_2 \gamma$ , *NPD* window size, and probability weight are 0.6,  $-4.7$ , 8, and 5, respectively for the  $SVM_{Polynomial}$  model; 0.4,  $-10.7$ , 8, and 4, respectively, for the  $SVM_{RBF}$  model; and 0.7,  $-2.9$ , 9, and 4, respectively, for the  $SVM_{Sigmoid}$  model.

In Step 6, the ensemble probabilities of individual candidate cysteine pairs are calculated, and then the probability thresholds ( $Th_1=0.44$  and  $Th_2=0.329$  in this example) of disulfide connectivity are determined accordingly (Step 7). In Step 8, Gabow's algorithm is used to search the solution with maximum weighted perfect matching according to the weighted graph consisting of six cysteines in the input protein sequence. Finally, in Step 9, the thresholds that were determined in Step 6 are used to select cysteine bonding pairs for output. As shown in Fig. 2, among the three cysteine pairs ( $C_1-C_3$ ,  $C_2-C_5$  and  $C_4-C_6$ ) that were obtained in Step 8, only two pairs ( $C_1-C_3$  and  $C_2-C_5$ ) with a connectivity probability greater than the thresholds are output.

## 3. Experimental results

### 3.1. Protein chain classification

The dataset SPXC was randomly divided into 10 folds, which is exactly the same number of folds as the experiment that was performed by Cheng et al. [4], for conducting the experiment with the values of SVM parameters and *NPD* window size that were obtained from the dataset MART with 20-fold cross-validation. As shown in Table 1, comparisons of the predictive performance between the method that was proposed in this study and the method that was proposed by Cheng et al. [4] are summarized. As indicated in the table, the average sensitivity (87.7%), specificity (86.2%), and accuracy (86.7%) of the SVM model are better than the method presented in Cheng et al. [4] with sensitivity, specificity, and accuracy of 56%, 83%, and 74%, respectively. In this application, the ensemble SVM model has converged with the single-kernel SVM classifier.

**Table 1**

Protein chain classification results for dataset SPXC.

|                  | Sensitivity (%) | Specificity (%) | Accuracy (%) |
|------------------|-----------------|-----------------|--------------|
| Cheng et al. [4] | 56              | 83              | 74           |
| $SVM_{RBF}$      | 87.7            | 86.2            | 86.7         |
| Ensemble SVM     | 87.7            | 86.2            | 86.7         |

**Table 2**

Residue classification results for dataset MART.

|                 | Sensitivity (%) | Specificity (%) | Accuracy ( $Q_2$ ) (%) |
|-----------------|-----------------|-----------------|------------------------|
| Chen et al. [8] | 77              | 97              | 90                     |
| $SVM_{RBF}$     | 92.8            | 92.1            | 92.4                   |
| Ensemble SVM    | 92.8            | 92.1            | 92.4                   |

Notably, the SPXC dataset contains 897 positive sequences and 1650 negative sequences, which presents unbalanced positive and negative cases with the number of negative sequences at two times that of the positive sequences. As reported in Cheng et al. [4], the specificity (83%) is much higher than the sensitivity (56%), indicating that the model that was trained by unbalanced positive and negative cases might incur bias. In contrast, the sensitivity and specificity obtained by the SVM model are similar, indicating that the ensemble SVM model is effective in reducing the bias caused by unbalance positive and negative samples.

### 3.2. Residue classification

The overall predictive accuracy  $Q_2$  is evaluated as  $Q_2 = N_c/N_a$  with  $N_c$  and  $N_a$  indicating the total number of correctly predicted bonding state of cysteines and the total number of cysteines, respectively. The sensitivity measures the proportion of actual positives that were correctly identified as positives, which can be defined as:  $sensitivity = TP/(TP + FN)$ . In contrast, the specificity measures the proportion of actual negatives that were correctly identified as negatives, which can be defined as:  $specificity = TN/(TN + FP)$ , in which *TP* indicates the number of bonded cysteines that were correctly identified as bonded, *FN* represents the number of bonded cysteines that were incorrectly identified as unbonded, *TN* denotes the number of unbonded cysteines that were correctly identified as unbonded, and *FP* is the number of unbonded cysteines that were incorrectly identified as bonded.

To compare these results with those results that were reported in Chen et al. [8] and Cheng et al. [4], the datasets MART and SPX were used for conducting the experiment. The results of residue classification using different methods for the datasets MART and SPX were compared. Table 2 compares the accuracy, sensitivity and specificity tested with the dataset MART with the methods that were proposed by Chen et al. [8] and this study. As shown in the table, the specificity achieves as high as 97%, whereas the sensitivity is only 77% for the method that was proposed by Chen et al. [8]. This result indicates that the model achieved greater predictions for the unbonded cysteines, but diminished in the predictions of bonded cysteines. In contrast, the sensitivity and specificity that were obtained in this study are 92.8% and 92.1%, respectively, indicating that our method balances the weights between bonded and unbonded cysteine prediction. Moreover, the predictive accuracy of our method (92.4% vs. 90%) outperforms the method that was proposed by Chen et al. [8].

For the dataset SPX, the accuracy, sensitivity and specificity between the methods proposed in this study and by Cheng et al. [4] are compared in Table 3. As indicated in the table, although the

specificity of this study achieves only 82.2%, which is smaller than the specificity (87.9%) that was obtained by Cheng et al. [4]. the sensitivity (95.7%) and accuracy (93.7%) that were obtained in this study are much higher than the results (sensitivity: 88.9% and accuracy: 88.7%) that were reported in Cheng et al. [4].

As mentioned in Section 2.3, if the Euclidean distance ( $D_{pi,pj}$ ) between two cysteines that are located at  $P_i$  and  $P_j$  is greater than 15, then they are deemed as unbonded without being connected by a disulfide bond. With this assumption, the number of candidates of oxidized cysteine pairs may decrease, thereby compromising the specificity but improving the sensitivity and accuracy of this method.

### 3.3. Bridge classification and the prediction of the disulfide bonding pattern without a prior knowledge of the bonding state of cysteines

To evaluate the predictive performance and measure the accuracy, two indexes were used, which were defined as  $Q_p = C_p/T_p$  and  $Q_c = C_c/T_c$ , in which  $C_p$  denotes the number of proteins whose

bonding patterns are correctly predicted;  $T_p$  is the total number of proteins in the test set;  $C_c$  represents number of disulfide bridges that are correctly predicted; and  $T_c$  indicates the total number of disulfide bridges in test proteins.

Tables 4 and 5 show the performance of the bridge classification and the disulfide bonding pattern prediction, respectively. As shown in these tables, the performance that was achieved with ensemble SVM ( $SVM_{Ensemble}$ ), which was based on the dataset SPX, are compared with the results that were obtained with a single-kernel SVM ( $SVM_{RBF}$ ) and that were reported in Cheng et al. [4]. As indicated in Table 4, only the sensitivity and specificity of the bridge classification against different numbers of bonds ( $B$ ) were reported, whereas the accuracy was not shown in the work that was performed by Cheng et al. [4]. According to the definition, the value of accuracy, which generally lies between sensitivity and specificity, can be approximately derived. It can be observed that without a prior knowledge of the bonding state of cysteines, except  $B=1$  (sensitivity=71%) and  $B=14$  (sensitivity=79%), the sensitivities are all lower than 70% for the rest of the cases, with an average sensitivity achieving only 52%. In contrast, for the ensemble SVM model, the accuracies are all greater than 91.7% for the number of bonds that were less than 26 ( $B=1-25$ ). Notably, there is only one protein sequence with 26 cysteines bonding pairs in the dataset SPX. In the experiment, the overall accuracy is considered the fitness. Although the accuracy of the bridge classification for the case with  $B=26$  is only 46.2%, i.e., 12 cysteine pairs have been predicted correctly, the overall mean predictive accuracy is as high as 96.5%, which is better than the method that was

**Table 3**  
Residue classification results for dataset SPX.

|                  | Sensitivity (%) | Specificity (%) | Accuracy ( $Q_c$ ) (%) |
|------------------|-----------------|-----------------|------------------------|
| Cheng et al. [4] | 88.9            | 87.9            | 88.7                   |
| $SVM_{RBF}$      | 95.7            | 82.2            | 93.7                   |
| Ensemble SVM     | 95.7            | 82.2            | 93.7                   |

**Table 4**  
Bridge classification results for dataset SPX without a prior knowledge of the bonding state of cysteines.

| # of bonds | # of sequence | Cheng et al. [4] |                 | $SVM_{RBF}$            | $SVM_{Ensemble}$       |
|------------|---------------|------------------|-----------------|------------------------|------------------------|
|            |               | Sensitivity (%)  | Specificity (%) | Accuracy ( $Q_c$ ) (%) | Accuracy ( $Q_c$ ) (%) |
| 1          | 398           | 71               | 48              | 82.2                   | 93.5                   |
| 2          | 211           | 59               | 60              | 93.1                   | 96.7                   |
| 3          | 219           | 55               | 61              | 96.3                   | 97.6                   |
| 4          | 88            | 44               | 48              | 98.9                   | 96.9                   |
| 5          | 49            | 32               | 35              | 99.2                   | 99.6                   |
| 6          | 19            | 32               | 36              | 98.2                   | 99.1                   |
| 7          | 10            | 29               | 32              | 100.0                  | 100.0                  |
| 8          | 10            | 20               | 22              | 100.0                  | 98.8                   |
| 9          | 2             | 44               | 52              | 100.0                  | 100.0                  |
| 10         | 3             | 33               | 36              | 100.0                  | 100.0                  |
| 12         | 2             | 38               | 39              | 91.7                   | 91.7                   |
| 14         | 1             | 79               | 85              | 100.0                  | 100.0                  |
| 16         | 2             | 13               | 13              | 93.8                   | 93.8                   |
| 17         | 2             | 53               | 60              | 94.1                   | 100.0                  |
| 25         | 1             | 32               | 53              | 100.0                  | 100.0                  |
| 26         | 1             | 31               | 51              | 100.0                  | 46.2                   |
| Overall    | 1018          | 52               | 51              | 94.6                   | 96.5                   |

**Table 5**  
Predictive accuracy of disulfide connectivity for dataset SPX without a prior knowledge of the bonding state of cysteines.

| # of bond | # of sequence | Cheng et al. [4]         |                            | $SVM_{RBF}$            | $SVM_{Ensemble}$       |
|-----------|---------------|--------------------------|----------------------------|------------------------|------------------------|
|           |               | $Q_p$ with SS and SA (%) | $Q_p$ with PSS and PSA (%) | Accuracy ( $Q_p$ ) (%) | Accuracy ( $Q_p$ ) (%) |
| 1         | 398           | 59                       | 59                         | 74.4                   | 82.4                   |
| 2         | 211           | 59                       | 56                         | 86.7                   | 93.8                   |
| 3         | 219           | 50                       | 47                         | 91.3                   | 94.1                   |
| 4         | 88            | 34                       | 22                         | 96.6                   | 92.0                   |
| 5–26      | 102           | 20                       | 13                         | 93.1                   | 93.1                   |
| Overall   | 1018          | 51                       | 48                         | 84.4                   | 89.2                   |

proposed by Cheng et al. [4] (51–52% or so) and the single-kernel SVM model (94.6%).

As compared in Table 5, regarding the prediction of disulfide connectivity, the predictive accuracy that was obtained by Cheng et al. [4] declines when the number of bonds is greater than 3, resulting in a mean accuracy of only 51%. In contrast, for both single-kernel and ensemble SVM models, although the number of bonds has been increased to a number greater than 3, the high predictive accuracy of both models remains. The mean predictive accuracy of the ensemble SVM model reaches 89.2%, which is 38.2% and 1.9% greater than that obtained by Cheng et al. [4] and that achieved by the single SVM model, respectively. As manifested in Tables 4 and 5, the ensemble SVM model, consisting of three SVM classifiers with individual kernels, outperforms a model that was built with a single-kernel SVM classifier, which is consistent with previous investigations [27–30].

#### 4. Discussion and conclusions

Recently, the predictive rate of the oxidation states of cysteines in protein sequences has improved significantly. However, the predictive rate of the disulfide bonding patterns of cysteines remains unsatisfactory. Nevertheless, much effort is required to improve the prediction accuracies. To our best knowledge, all the previous approaches except the one presented by Cheng et al. [4] assume that the oxidation states of cysteines are known in advance. To practically solve the problem in the prediction of disulfide bonding pattern, this assumption should be removed. In this study, without a prior knowledge of the oxidation states of cysteines, the integrated MTS and ensemble SVM method that were used for fine-tuning the window size of *NPD*, weight of probability, and SVM parameters achieves the accuracy of 89.2% in the prediction of disulfide bonding patterns, which greatly outperforms the single-kernel SVM and the method that was proposed by Cheng et al. [4].

Previous studies generally used features that were obtained from a single cysteine with local characteristics. In contrast, in this study, protein structure was directly considered to obtain the *NPD*, which was used as the feature representing more global characteristics of the protein, that might be the key factor in increasing predictive performance. In addition, the ensemble SVM model that was used for predicting the bonding probability of individual cysteine pairs further increases the predictive performance.

Although a single-kernel SVM model can reach good performance, Dietterich [30] demonstrated that the performance of the ensemble classifiers might outperform a single model. The reasons for such an improved performance might be: (1) if several different optimal assumptions exist, then the ensemble classifiers can reduce the risk of choosing a wrong assumption; (2) a single machine learning algorithm may result in obtaining local optima, whereas, the ensemble method may obtain a better solution; and (3) in general, a single model cannot achieve the desired function.

A general survey of ensemble learning and a theoretical description of why ensemble learning may outperform the single model were also addressed by Brown et al. [33]. The methods were categorized into three strategies, including changing the starting point in the hypothesis space, training sets of accessible hypotheses, and hypothesis space traversal, to achieve diversity. Changing the starting points within the hypothesis space by creating different initial settings for achieving models with better performance was suggested. In addition, a better model can be obtained by varying the training sets that are accessible by the employed ensemble method, such as bagging, boosting, and random sub-space [33].

The modified maximum weight perfect matching method, which was accompanied by two probability thresholds, was used to effectively determine whether a bond exists between two cysteines. This method greatly facilitates Gabow's algorithm to be suitable for the prediction of disulfide bonding patterns directly from the protein sequences without a prior knowledge of the bonding state of cysteines. In addition, this method can be used to predict the disulfide bonding patterns without requiring two stages of processing as suggested in previous investigations with the first stage used to predict the bonding state of cysteines, which are then used in the second stage to predict the disulfide bonding pattern.

For protein chain and residue classifications, the probability weights of the ensemble SVM model that were obtained in the training phase are 0, 1, and 0 for  $SVM_{Sigmoid}$ ,  $SVM_{RBF}$  and  $SVM_{Polynomial}$ , respectively, indicating that it is possible for an ensemble SVM model to converge to a single-kernel  $SVM_{RBF}$  model in some applications. For applications of bridge classification and disulfide bonding pattern prediction, the ensemble SVM model still outperforms the single-kernel  $SVM_{RBF}$  model.

In conclusion, the disulfide bonding pattern is directly predicted with the feature *NPD* that was obtained from individual pairs of cysteines in our proposed method, instead of first predicting the bonding state of each cysteine, followed by the prediction of the bonding pattern that was adopted in the traditional methods. Our proposed method, which was based on the ensemble SVM classifier outperforms the methods that were proposed in previous studies with improved predictive accuracy.

#### Conflict of interest statement

None declared.

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#### References

- [1] J. Skolnick, A. Kolinski, A.R. Ortiz, MONSSTER: a method for folding globular proteins with a small number of distance restraints, *J. Mol. Biol.* 265 (1997) 217–241.
- [2] E.S. Huang, R. Samudrala, J.W. Ponder, Ab initio fold prediction of small helical proteins using distance geometry and knowledge-based scoring functions, *J. Mol. Biol.* 290 (1999) 267–281.
- [3] S. Niu, T. Huang, K. Feng, et al., Inter- and intra-chain disulfide bond prediction based on optimal feature selection, *Protein Pept. Lett.* 20 (2013) 324–335.
- [4] J. Cheng, H. Saigo, P. Baldi, Large-scale prediction of disulphide bridges using Kernel methods, two-dimensional recursive neural networks, and weighted graph matching, *Proteins* 62 (2006) 617–629.
- [5] A. Fiser, I. Simon, Predicting the oxidation state of cysteines by multiple sequence alignment, *Bioinformatics* 16 (2000) 251–256.
- [6] P. Fariselli, P. Riccobelli, R. Casadio, Role of evolutionary information in predicting the disulfide-bonding state of cysteine in proteins, *Proteins* 36 (1999) 340–346.
- [7] P.L. Martelli, P. Fariselli, L. Malaguti, R. Casadio, Prediction of the disulfide-bonding state of cysteines in proteins with hidden neural networks, *Protein Eng.* 15 (2002) 951–953.

- [8] Y.C. Chen, Y.S. Lin, C.J. Lin, J.K. Hwang, Prediction of the bonding states of cysteines using the support vector machines based on multiple feature vectors and cysteine state sequences, *Proteins* 55 (2004) 1036–1042.
- [9] P. Fariselli, R. Casadio, Prediction of disulfide connectivity in proteins, *Bioinformatics* 17 (2001) 957–964.
- [10] A. Vullo, P. Frasconi, Disulfide connectivity prediction using recursive neural networks and evolutionary information, *Bioinformatics* 20 (2004) 653–659.
- [11] F. Ferré, P. Clote, Disulfide connectivity prediction using secondary structure information and diresidue frequencies, *Bioinformatics* 21 (2005) 2336–2346.
- [12] C.H. Tsai, B.J. Chen, H.H. Chan, et al., Improving disulfide connectivity prediction with sequential distance between oxidized cysteines, *Bioinformatics* 21 (2005) 4416–4419.
- [13] Y.C. Chen, J.K. Hwang, Prediction of disulfide connectivity from protein sequences, *Proteins* 61 (2005) 507–512.
- [14] E. Zhao, H.L. Liu, C.H. Tsai, et al., Cysteine separations profiles on protein sequences infer disulfide connectivity, *Bioinformatics* 21 (2005) 1415–1420.
- [15] B.J. Chen, C.H. Tsai, C.H. Chan, C.Y. Kao, Disulfide connectivity prediction with 70% accuracy using two-level models, *Proteins* 64 (2006) 246–252.
- [16] C.H. Lu, Y.C. Chen, C.S. Yu, J.K. Hwang, Predicting disulfide connectivity patterns, *Proteins* 67 (2007) 262–270.
- [17] J. Song, Z. Yuan, H. Tan, et al., Predicting disulfide connectivity from protein sequence using multiple sequence feature vectors and secondary structure, *Bioinformatics* 23 (2007) 3147–3154.
- [18] H.H. Lin, L.Y. Tseng, Prediction of disulfide bonding pattern based on a support vector machine and multiple trajectory search, *Inf. Sci.* 199 (2012) 167–178.
- [19] S.M. Marino, V.N. Gladyshev, Analysis and functional prediction of reactivity cysteine residues, *J. Biol. Chem.* 287 (2012) 4419–4425.
- [20] A. Sali, T.L. Blundell, Comparative protein modelling by satisfaction of spatial restraints, *J. Mol. Biol.* 234 (1993) 779–815.
- [21] H.H. Lin, L.Y. Tseng, DBCP: a web server for disulfide bonding connectivity pattern prediction without the prior knowledge of the bonding state of cysteines, *Nucleic Acids Res.* 38 (2010) W503–W507.
- [22] H.M. Berman, T. Battistuz, T.N. Bhat, et al., The Protein Data Bank, *Acta Crystallogr. D* 58 (2002) 899–907.
- [23] S. Mika, B. Rost, UniqueProt: creating representative protein sequence sets, *Nucleic Acids Res.* 31 (2003) 3789–3791.
- [24] C. Sander, R. Schneider, Database of homology-derived protein structures and the structural meaning of sequence alignment, *Proteins* 9 (1991) 56–68.
- [25] L.Y. Tseng, C. Chen, Multiple trajectory search for large scale global optimization, in: *Proceedings of the 2008 IEEE Congress on Evolutionary Computation, CEC 2008*, IEEE, 2008, pp. 3057–3064.
- [26] L.Y. Tseng, C. Chen, Multiple trajectory search for multiobjective optimization, in: *Proceedings of the 2007 IEEE Congress on Evolutionary Computation, CEC 2007*, IEEE, 2007, pp. 3609–3616.
- [27] T. Sun, L. Jiao, F. Liu, S. Wang, J. Feng, Selective multiple kernel learning for classification with ensemble strategy, *Pattern Recognition* 46 (2013) 3081–3090.
- [28] L. Rokach, Ensemble-based classifiers, *Artif. Intell. Rev.* 33 (2010) 1–39.
- [29] Q. Li, B. Yang, Y. Li, N. Deng, L. Jing, Constructing support vector machine ensemble with segmentation for imbalanced datasets, *Neural Comput. Appl.* 22 (2013) S249–S256.
- [30] T.G. Dietterich, Ensemble methods in machine learning, in: *Proceedings of the 1st International Workshop on Multiple Classifier Systems*, 2000, pp. 1–15.
- [31] C.C. Chang, C.J. Lin, LIBSVM: a library for support vector machines, *ACM Trans. Intell. Syst. Technol.* 2 (2011) 1–27.
- [32] H.N. Gabow, Implementation of algorithms for maximum matching on nonbipartite graphs (Ph.D. thesis), Stanford University, 1973.
- [33] G. Brown, J. Wyatt, R. Harris, X. Yao, Diversity creation methods: a survey and categorisation, *Inf. Fusion* 6 (1) (2005) 5–20.